

Application Design



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Application design: a continuous process



The Mock



Providing houses all over the world

Houses currently on the market

Address	Country	Asking Price
12 Valley of Kings, Geneva	Switzerland	\$900,000
89 Road of Forks, Bern	Switzerland	\$500,000



Technology

Vite

Next.js



Design Decisions

No hard rules

Often designer/team decision

Unwanted: components that do too much

Design is a living thing

Extracting components part of the process



The Mock: House Composition



Providing houses all over the world



Switzerland

89 Road of Forks, Bern

\$500,000

This impressive family home, which dates back to approximately 1840, offers original period features throughout and is set back from the road with off street parking for up to six cars and an original Coach House, which has been incorporated into the main house to provide further accommodation.

Bidder	Amount
Mohammed Valis	\$302,100
Jane Williams	\$310,500
John Kepler	\$315,100

Bidder

0

Add



1. Break up Into a Component Hierarchy

- App
 - Banner
 - HouseList
 - HouseRow
 - House
 - BidList
 - AddBid



File Structure

components

app.js

houseList.js

house.js

...

helpers

currencyFormatter.js

loadingStatus.js

navValues.js

...

hooks

useHouses.js

useBids.js

useGetRequest.js



File Structure

app.js

houses

houseList.js

house.js

useHouses.js

useBids.js

...

customers

customerList.js

useCustomers.js

...

common

currencyFormatter.js

loadingStatus.js

navValues.js

useGetRequest.js

...



File Structure

app.js

houses

index.js



import HouseList from './houses'

house.js

useHouses.js

useBids.js

...

customers

index.js



import CustomerList from './customers'

useCustomers.js

...

common

currencyFormatter.js

loadingStatus.js

navValues.js

useGetRequest.js

...



Determining a File Structure

React doesn't have an opinion

Choose something that works for the application at hand

Limit to 3 or 4 directory levels



Create a First Application Version

No interactivity support

Without state

Just props

In-memory data source



When Application Data is a State Candidate

It is **not** passed in
by the parent via a
prop

It changes over
time

It can **not** be
computed



Determining Application State

Houses

Yes

House count

No

Bids

Yes

New Bid

Yes



**There is only one component
that owns the state**



Placing State

- App
 - Banner
 - HouseList
 - HouseRow
 - House
 - BidList
 - AddBid



How to Apply State?

useState

Custom hook

Context



Applying State

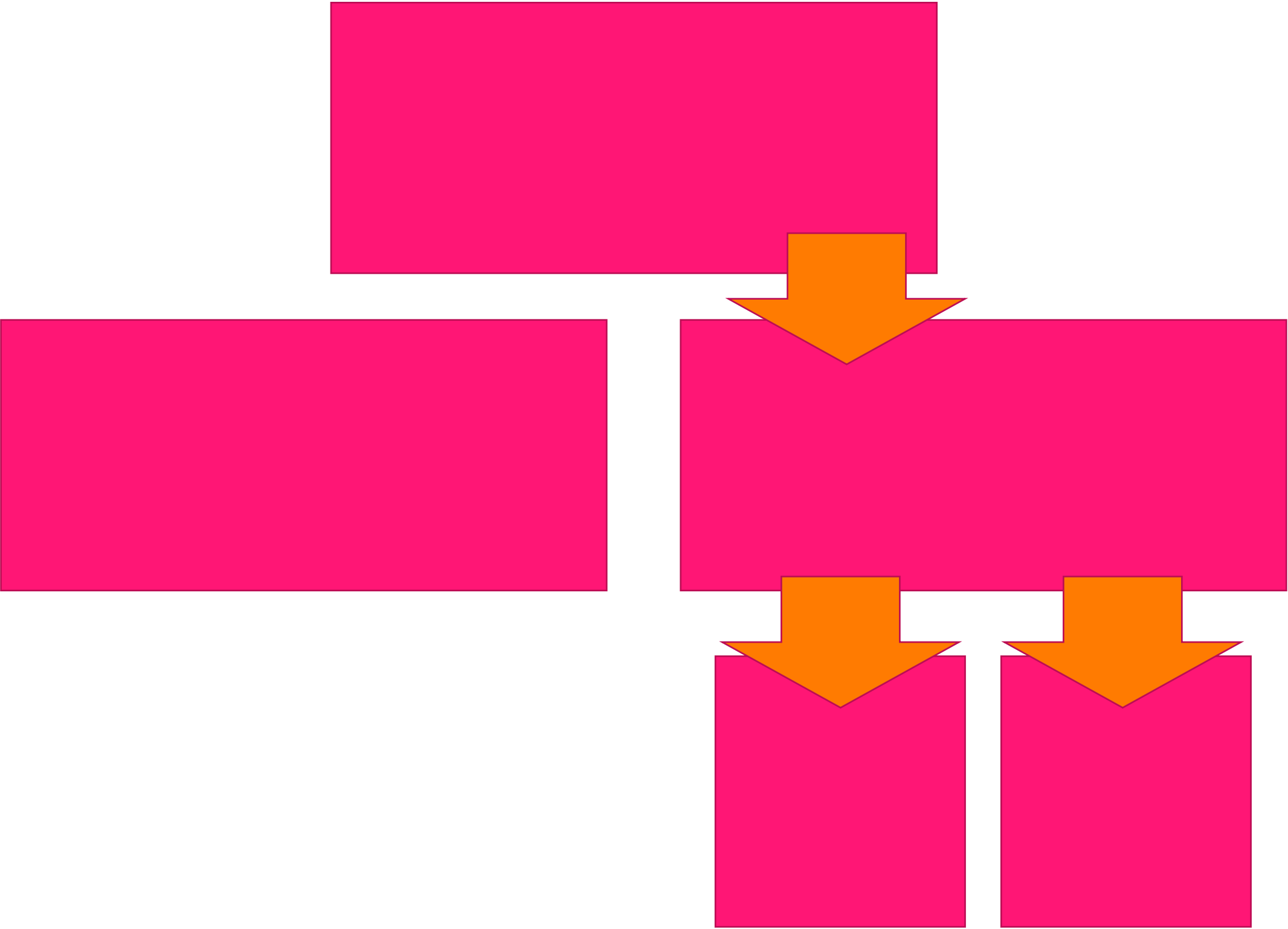
What

Where

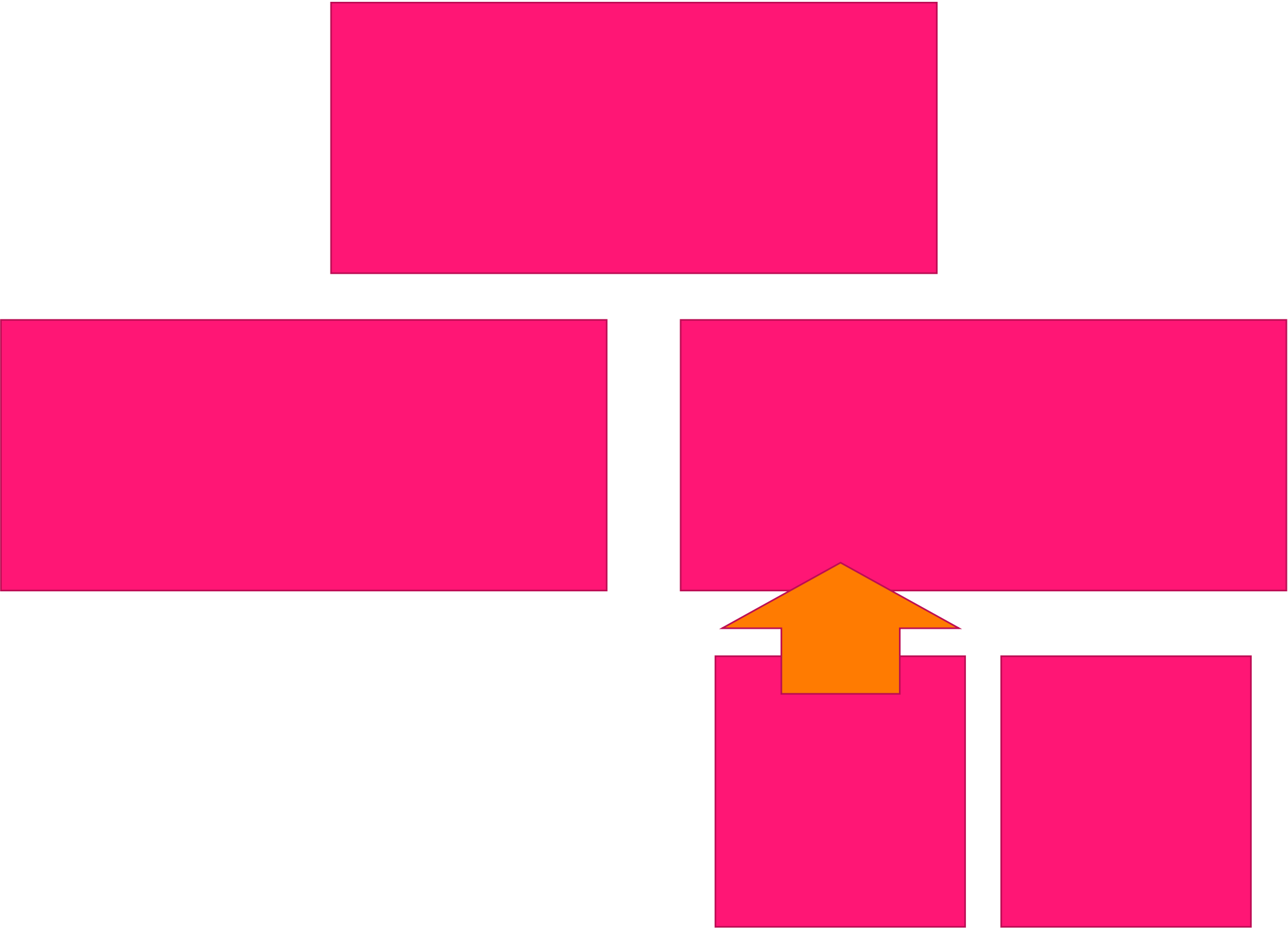
How



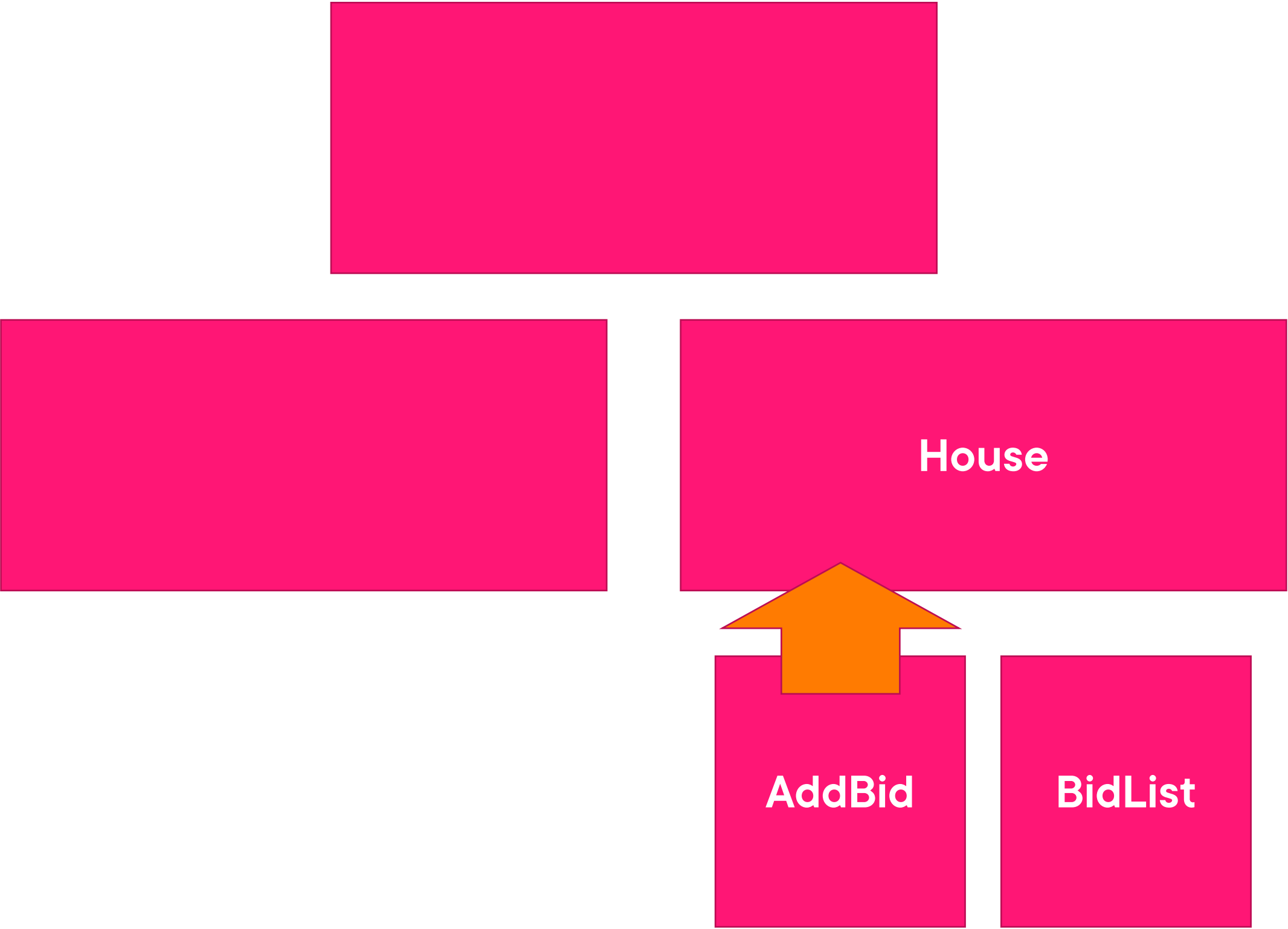
React's Data Flow



Inversing Data Flow



Inversing Data Flow



Implementing Design Changes

- App
 - Banner
 - HouseList
 - HouseRow
 - House
 - BidList
 - AddBid



Where to Go Next

Practice fundamentals first

Create an application by yourself

Design first

Advanced courses



Related Topics

TypeScript

Redux

TanStack React Query



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